DAY 5: CONFIGURING OSPF

Multi-Area OSPF and Reconvergence Practical Report

Objective

To configure OSPF across two areas using three routers, connect two PCs in different areas, verify OSPF neighbor relationships and end-to-end connectivity, and observe routing changes when the connection between Area 1 and the backbone Area 0 fails and is restored.

Theory

Open Shortest Path First (OSPF) is a link-state routing protocol that calculates routes using the lowest total interface cost. OSPF supports multiple areas to organize a large routing domain. Area 0 is the backbone area. Other areas should connect to Area 0 through an Area Border Router (ABR). An ABR has interfaces in more than one OSPF area and shares appropriate routing information between them.

OSPF reconvergence is the process in which routers detect a topology change, update their link-state databases, run the SPF calculation when needed, and update routing tables. Recovery time depends on the type of failure, OSPF timers, topology, and Packet Tracer simulation behavior.

Topology and Addressing Plan

Use three routers in a line: PC1 — R1 — R2 — R3 — PC2. R1 and R2 form Area 0; the R2–R3 link and R3's PC-facing LAN are in Area 1. R2 is the Area Border Router (ABR).

Device	Interface	IP address	Subnet mask	OSPF area
PC1	Ethernet	192.168.10.10	255.255.255.0	—
R1	G0/0 (to PC1)	192.168.10.1	255.255.255.0	Area 0
R1	G0/1 (to R2)	10.0.12.1	255.255.255.252	Area 0
R2	G0/0 (to R1)	10.0.12.2	255.255.255.252	Area 0
R2	G0/1 (to R3)	10.0.23.1	255.255.255.252	Area 1
R3	G0/0 (to R2)	10.0.23.2	255.255.255.252	Area 1
R3	G0/1 (to PC2)	192.168.20.1	255.255.255.0	Area 1
PC2	Ethernet	192.168.20.10	255.255.255.0	—

Default gateways: PC1 = 192.168.10.1; PC2 = 192.168.20.1. Interface names may differ by router model; substitute the actual interface names in Packet Tracer. This is a single-path topology, so shutting the R2–R3 link will interrupt PC-to-PC connectivity until the link is restored; it does not have an alternate route around the failure.

Step 1: Build the Network

1. Open Cisco Packet Tracer and place three routers, two PCs, and the required Ethernet cables.
2. Connect PC1 to R1, R1 to R2, R2 to R3, and R3 to PC2.
3. Configure the IP addresses and masks shown in the addressing table. Set each router interface to no shutdown.
4. Configure each PC's IP address and default gateway using the Desktop > IP Configuration window.

5. Verify the local links with show ip interface brief on routers and ping the directly connected gateway from each PC.

Step 2: Configure R1 (Area 0)

```
enable
configure terminal
hostname R1
interface gigabitEthernet0/0
 ip address 192.168.10.1 255.255.255.0
 no shutdown
exit
interface gigabitEthernet0/1
 ip address 10.0.12.1 255.255.255.252
 no shutdown
exit
router ospf 1
 router-id 1.1.1.1
 network 192.168.10.0 0.0.0.255 area 0
 network 10.0.12.0 0.0.0.3 area 0
end
write memory
```

Step 3: Configure R2 (ABR: Area 0 and Area 1)

```
enable
configure terminal
hostname R2
interface gigabitEthernet0/0
 ip address 10.0.12.2 255.255.255.252
 no shutdown
exit
interface gigabitEthernet0/1
 ip address 10.0.23.1 255.255.255.252
 no shutdown
exit
router ospf 1
 router-id 2.2.2.2
 network 10.0.12.0 0.0.0.3 area 0
 network 10.0.23.0 0.0.0.3 area 1
end
write memory
```

Step 4: Configure R3 (Area 1)

```
enable
configure terminal
hostname R3
interface gigabitEthernet0/0
 ip address 10.0.23.2 255.255.255.252
 no shutdown
exit
interface gigabitEthernet0/1
 ip address 192.168.20.1 255.255.255.0
 no shutdown
exit
router ospf 1
 router-id 3.3.3.3
 network 10.0.23.0 0.0.0.3 area 1
 network 192.168.20.0 0.0.0.255 area 1
end
write memory
```

Step 5: Verify OSPF Neighbors and Routes

- **show ip interface brief —** Confirm interfaces are up/up.
- **show ip ospf neighbor —** R1 should see R2; R2 should see R1 and R3; R3 should see R2 after adjacency forms.
- **show ip ospf —** Review OSPF process, router ID, areas, and protocol information.
- **show ip route —** Look for OSPF-learned routes marked with O or O IA.
- **show ip route ospf —** Display routes learned through OSPF.
- **ping 192.168.20.10 —** From PC1, ping PC2. Also ping 192.168.10.10 from PC2.

Note: A neighbor relationship forms only between routers sharing a link and matching OSPF settings. R1 and R3 are not direct neighbors in this topology. On Ethernet links, fully formed neighbors often show FULL/DR or FULL/BDR; the exact state depends on network type.

Step 6: Simulate Link Failure

The task asks to simulate a router failure by shutting down the interface connecting Area 1 to Area 0. In this topology, use R2's G0/1 interface (10.0.23.1) to bring down the R2–R3 link. This models a link failure rather than powering off an entire router.

```
R2# configure terminal
R2(config)# interface gigabitEthernet0/1
R2(config-if)# shutdown
R2(config-if)# end
R2# show ip ospf neighbor
R2# show ip route
R2# show ip ospf
```

Immediately test a ping from PC1 to PC2 and record whether it fails. Check the neighbor list and routing table on R2 and R3. The R2–R3 adjacency should go away and routes to the remote Area 1 LAN should no longer be usable through that link. Because this example has no redundant path, connectivity is expected to remain unavailable while the link is down.

Step 7: Restore the Link and Observe Recovery

```
R2# configure terminal
R2(config)# interface gigabitEthernet0/1
R2(config-if)# no shutdown
R2(config-if)# end
R2# show ip ospf neighbor
R2# show ip route
R2# show ip ospf
```

Wait for the R2–R3 adjacency to return to FULL, then ping PC2 from PC1 again. The OSPF routes should reappear after adjacency and route calculation complete. Use Packet Tracer Simulation mode or a timer to record approximate failure-detection and recovery times; actual values can vary.

Observation Table

Stage	Neighbor / routing observation	PC1-to-PC2 ping	Time observed
Normal operation	Record neighbors and OSPF routes before failure.	Success / loss: _____	_____ seconds
After shutdown	Record adjacency loss and route changes.	Success / loss: _____	_____ seconds
After no shutdown	Record adjacency recovery and routes returning.	Success / loss: _____	_____ seconds

Troubleshooting

- If a neighbor does not appear, check interface status, IP addressing, OSPF area numbers on the shared link, and router IDs.
- If a PC cannot ping its gateway, check the PC IP address, subnet mask, default gateway, cable, and router interface.
- If the OSPF routes do not appear, check the network statements with show ip protocols and verify the interfaces belong to the intended areas.
- If the topology does not recover, confirm the interface is enabled with no shutdown and wait for OSPF adjacency to reform.

Conclusion

This practical demonstrates multi-area OSPF with Area 0 as the backbone and R2 acting as an Area Border Router between Area 0 and Area 1. OSPF neighbor and routing commands are used to verify normal operation.

Shutting down and restoring the R2–R3 link demonstrates how a topology change removes and later restores routes. Complete the observation table with the actual Packet Tracer results and measured times.